Practical - 9

Practical Title: Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

Problem Definition: Implement secure user registration by inserting user details into MySQL using MYSQLi. Validate input on frontend and backend, check duplicate email/username, hash passwords using password_hash(), and display appropriate messages.

A. Theory

1. Is password_hash() used before storing passwords?

Answer: Yes, password_hash($password, PASSWORD_DEFAULT) hashes passwords before SQL insertion.

2. Is duplicate email/username checked?

Answer: Yes, a prepared SELECT query checks for existing emails before inserting.

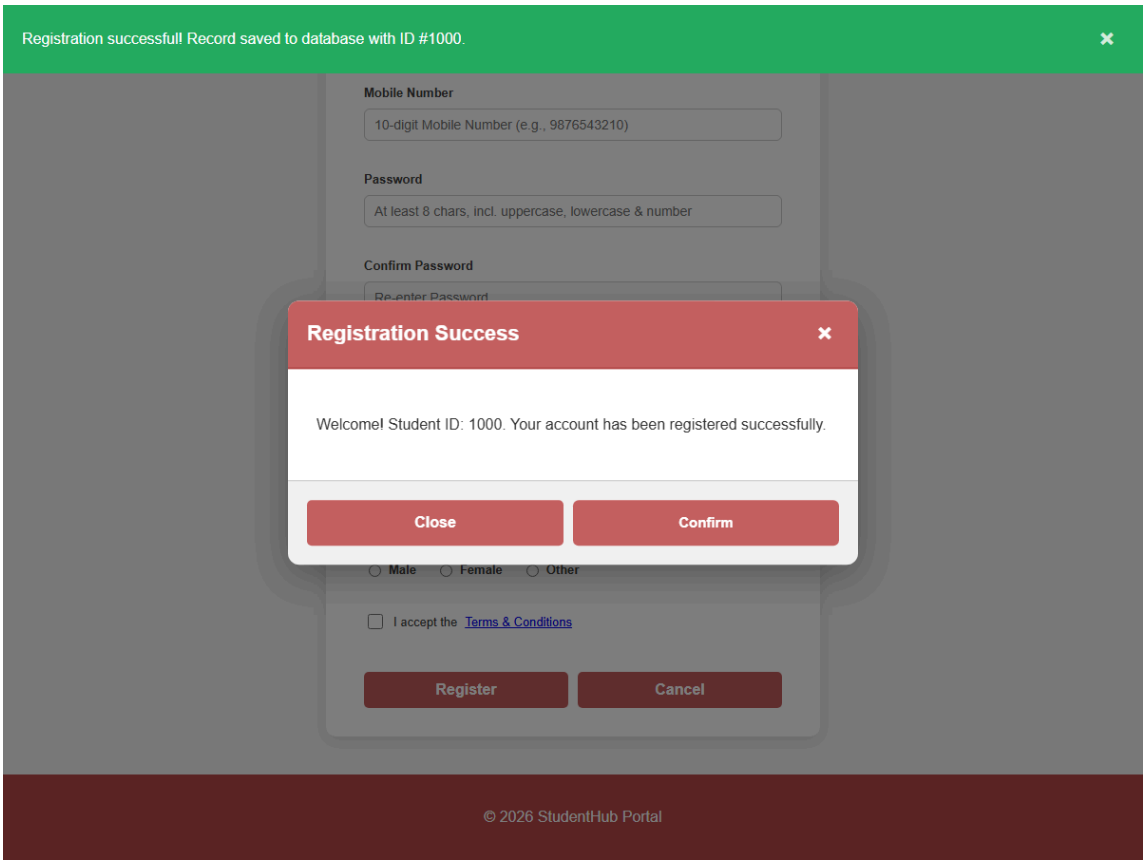
3. Are frontend and backend validations consistent?

Answer: Yes, required fields, email formats, and password lengths match across JS and PHP.

B. Implementation/ Output/Result

Feature	Implementation
Database Connection	MySQLi object (new mysqli())
Validation	Frontend (JS/HTML) and backend (filter_var, empty())
Duplicate email check	MySQLi prepared statement (SELECT student_id ... WHERE email = ?)
Password Security	One-way BRYPT hashing using password_hash()
Data Insertion	Parameterized query via $stmt->bind_param() & execute()

C. Evidence of Work (Screenshots)



Description: Figure 9.1 registration success

D. GitHub Repository Link:

Link: https://github.com/prep04-yl/StudentHub-Portal

E. Challenges Faced:

Handling duplicate email conditions while ensuring the password hash isn't truncated by an undersized database column.

F. Learning Outcome:

I learned how to implement secure user registration following basic web security practices.

Faculty Remarks: (Keep this blank for faculty)